

Chengyin Li

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Appointment	Radiation Oncology Researcher/Programmer Department of Radiation Oncology Henry Ford Health Detroit, MI, USA	Aug. 2024 - Present
Research Interests	Medical Image Analysis with Deep Learning • Automatic medical image segmentation • Multi-modal learning • Foundation models in medical domain • Clinical-oriented AI solutions for healthcare Trustworthy AI • Fairness, Explainability, Robustness	
Education	Wayne State University, Detroit, Michigan, USA • Ph.D. in Computer Science • Advisor: Prof. Dongxiao Zhu • Dissertation: Novel Transformer Architectures for 3D Multi-Modal and Multi-Organ Medical Image Segmentation University of Chinese Academy of Sciences, Beijing, China • M.E. in Chemical Engineering Nanjing University of Science and Technology, Nanjing, China • B.E. in Chemical Engineering	Sept. 2019 - June 2024 Sept. 2013 - July 2016 Sept. 2009 - July 2013
Work Experience	Department of Radiation Oncology, Henry Ford Health • Research Scientist (part-time), 2D/3D Medical Image Analysis • Mentor: Dr. Indrin J. Chetty Trustworthy AI Lab, Wayne State University • Graduate Research Assistant, Deep Learning and Trustworthy AI • Supervisor: Prof. Dongxiao Zhu The Shenzhen Institutes of Advanced Technology • Visiting Student (part-time), Low-level Computer Vision • Mentor: Prof. Yu Qiao The Chinese University of Hong Kong (SZ) • Ph.D. Student, AI for Chemistry • Graduate Teaching Assistant in Computer and Information Engineering	May 2022 - July 2024 Sept. 2019 - May 2024 Sept. 2017 - Dec. 2018 Sept. 2017 - Sept. 2018

CVPR, ECCV, WACV, IJCAI, NeurIPS, and MICCAI are among the leading conferences in computer vision, artificial intelligence, machine learning, and medical imaging, with Google Scholar Metrics reporting h5-index scores of 450 for CVPR, 371 for NeurIPS, 262 for ECCV, 131 for WACV, 136 for IJCAI, and 96 for MICCAI—NeurIPS is ranked 1st among AI journals and conferences in engineering and computer science and seventh overall. Additionally, top journals such as Transactions on Machine Learning Research (TMLR), Medical Physics (h5-index 81) in radiology and medical imaging, and Pediatric Research (h5-index 64) in pediatric medicine further highlight the quality of research in these domains. Current impact metrics (June 29, 2026): 599 citations, h-index: 10, i10-index: 12 by  [Google Scholar](#).

Peer-Reviewed Conferences/Proceedings

- [C1] Rafi Ibn Sultan, Hui Zhu, Xiao Zhou, Chengyin Li, Prashant Khanduri, Marco Brocanelli, and Dongxiao Zhu. WalkGPT: Grounded Vision-Language Conversation with Depth-Aware Segmentation for Pedestrian Navigation. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
- [C2] Rafi Ibn Sultan, Chengyin Li, Hui Zhu, Prashant Khanduri, Marco Brocanelli, and Dongxiao Zhu. GeoSAM: Fine-tuning SAM with Sparse and Dense Visual Prompting for Automated Segmentation of Mobility Infrastructure. The 28th European Conference on Artificial Intelligence (ECAI), 2025.
- [C3] Chengyin Li, Doris Rusu, Jennifer Dolan, Parag J. Parikh, Kundan Thind. Leveraging Prior Fractions to Improve Segmentation in Fractionated Adaptive Radiotherapy. The 71st Annual Scientific Meeting (ASM), 2025. Oral.
- [C4] Chengyin Li, Hui Zhu, Rafi Ibn Sultan, Hassan Bagher Ebadian, Prashant Khanduri, Chetty Indrin, Kundan Thind, and Dongxiao Zhu. MulModSeg: Enhancing Unpaired Multi-Modal Medical Image Segmentation with Modality-Conditioned Text Embedding and Alternating Training. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
- [C5] Chengyin Li, Prashant Khanduri, Yao Qiang, Rafi Ibn Sultan, Indrin Chetty, and Dongxiao Zhu. AutoProSAM: Automated prompting SAM for 3D multi-organ segmentation. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025.
- [C6] Chengyin Li, Rafi Ibn Sultan, Hassan Bagher-Ebadian, Yao Qiang, Kundan Thind, Dongxiao Zhu, and Indrin Chetty. On the Implementation and Evaluation of Loss Functions for Robust Multiple Anatomy Segmentation on CT Images. International Conference on the use of Computers in Radiation Therapy (ICCR), 2024.
- [C7] Chengyin Li, Yao Qiang, Rafi Ibn Sultan, Hassan Bagher-Ebadian, Prashant Khanduri, Indrin J. Chetty, and Dongxiao Zhu. FocalUNETR: A Focal Transformer for Boundary-aware Prostate Segmentation using CT Images. International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023.
- [C8] Chengyin Li, Zheng Dong, Nathan Fisher, and Dongxiao Zhu. Coupling User Preference with External Rewards to Enable Driver-centered and Resource-aware EV Charging Recommendation. The 23rd European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD), Oral, 2022.
- [C9] Qiang, Yao, Chengyin Li, Prashant Khanduri, and Dongxiao Zhu. Interpretability-aware vision transformer. International Joint Conference on Neural Networks (IJCNN), 2025.
- [C10] Yao Qiang, Chengyin Li, Prashant Khanduri, Dongxiao Zhu. Fairness-aware Vision Transformer via Debaised Self-Attention. European Conference on Computer Vision (ECCV), 2024.
- [C11] Mohammad Peivandi, Jason Zhang, Michael Lu, Chengyin Li, Dongxiao Zhu, and Zhifeng Kou. Empirical Evaluation of the Segment Anything Model (SAM) for Brain Tumor Segmentation. IEEE International Symposium on Biomedical Imaging (ISBI), 2024.

- [C12] Xin Li, Deng Pan, Chengyin Li, Yao Qiang, and Dongxiao Zhu. Negative Flux Aggregation to Estimate Feature Attributions. The 32nd International Joint Conference on Artificial Intelligence (IJCAI), 2023.
- [C13] Yao Qiang, Chengyin Li, Marco Brocanelli, and Dongxiao Zhu. Counterfactual interpolation augmentation (CIA): A unified approach to enhance fairness and explainability of DNN. The 31st International Joint Conference on Artificial Intelligence (IJCAI), 2022.
- [C14] Yao Qiang, Deng Pan, Chengyin Li, Xin Li, Rhongho Jang, and Dongxiao Zhu. AttCAT: Explaining Transformers via Attentive Class Activation Tokens. The Thirty-Sixth Annual Conference on Neural Information Processing Systems (NeurIPS), 2022.
- [C15] Xin Li, Chengyin Li, and Dongxiao Zhu. COVID-MobileXpert: On-device COVID-19 Patient Triage and Follow-up Using Chest X-rays. International Conference on Bioinformatics and Biomedicine (BIBM), 2020.

Peer-Reviewed Journals

- [J1] Prashant Khanduri, Chengyin Li, Rafi Ibn Sultan, Yao Qiang, Joerg Kliewer, and Dongxiao Zhu. On Federated Compositional Optimization: Algorithms, Analysis, and Guarantees. Transactions on Machine Learning Research (TMLR), 2026.
- [J2] Chengyin Li, Rafi Ibn Sultan, Hassan Bagher-Ebadian, Yao Qiang, Kunda Thind, Dongxiao Zhu, and Indrin J. Chetty. Enhancing CT Image Segmentation Accuracy Through Ensemble Loss Function Optimization. Medical Physics, 2025.
- [J3] Chengyin Li, Hassan Bagher-Ebadian, Rafi Ibn Sultan, Mohamed Elshaikh, Benjamin Movsas, Dongxiao Zhu, and Indrin J. Chetty. A New Architecture Combining Convolutional and Transformer-Based Networks for Automatic 3D Multi-Organ Segmentation on CT Images. Medical Physics, 2023.
- [J4] Chengyin Li, Rhea E. Sullivan, Rhea E. Sullivan, Dongxiao Zhu, and Steven D. Hicks. Putting the ‘mi’ in Omics: Discovering Mirna Biomarkers for Pediatric Precision Care. Pediatric Research, 2022.

Peer-Reviewed Workshops

- [W1] Prashant Khanduri, Chengyin Li, Rafi Ibn Sultan, Yao Qiang, Joerg Kliewer, and Dongxiao Zhu. Proximal Compositional Optimization for Distributionally Robust Learning. ICML New Frontiers in Adversarial Machine Learning Workshop, 2023.
- [W2] Xin Li, Yao Qiang, Chengyin Li, Sijia Liu, and Dongxiao Zhu. Saliency-guided Adversarial Training for Learning Generalizable Features with Applications to Medical Imaging Classification System. ICML New Frontiers in Adversarial Machine Learning Workshop, 2022.

Peer-Reviewed Conference Abstracts

- [A1] Chengyin Li, Rafi Ibn Sultan, Doris N. Rusu, Hassan Bagher-Ebadian, Anthony J. Doemer, Dongxiao Zhu, Parag Parikh, and Kundan S. Thind. Disentangling Patient-Specific Canonical Anatomy and Deformation Manifolds for Improved MRI-Guided Adaptive Radiotherapy Segmentation. 2026 Joint AAPM|COMP Annual Meeting, Vancouver, BC (Snap Oral); to appear in Medical Physics, 2026.
- [A2] Chengyin Li, Doris N. Rusu, Jennifer L. Dolan, Parag J. Parikh, and Kundan S. Thind. Enhancing Adaptive Radiotherapy Segmentation with a 3D Unet Framework and Prior Fraction Information (SU-330-GPD-T-485). AAPM Annual Meeting; Medical Physics 52(10), e70059303–e70059304, 2025.
- [A3] Rafi Ibn Sultan, Chengyin Li, Ahmed I. Ghanem, Joon P. Kim, Hassan Bagher-Ebadian, Dongxiao Zhu, and Kundan S. Thind. NA-Unetr: A Neighborhood Attention Transformer Network for Enhanced 3D Segmentation of the Left Anterior Descending Artery (SU-330-GPD-T-688). AAPM Annual Meeting; Medical Physics 52(10), e70059368, 2025.

Pre-prints

- [P1] Rafi Ibn Sultan, Hui Zhu, Chengyin Li, and Dongxiao Zhu. BiPVL-Seg: Bidirectional Progressive Vision-Language Fusion with Global-Local Alignment for Medical Image Segmentation. arXiv:2503.23534, 2025.

Honors And Awards

IEEE Transactions on Medical Imaging (TMI) Distinguished Reviewer	2024-2025
Michael E. Conrad Award (Highest Honor at WSU CS Department)	2024
Outstanding Graduate Research Assistant Award	2024
Graduate Student Professional Travel Award	2022
Thomas C. Rumble Fellowship Award	Aug. 2019 - May 2020

Services

Editors/Co-editors

- JAIKE: Journal of Artificial Intelligence and Knowledge Engineering (Guest Editor) 2025

Program Committee Member/Conference Reviewer

- CVPR: IEEE/CVF Conference on Computer Vision and Pattern Recognition 2024-2026
- NeurIPS: Conference on Neural Information Processing Systems 2024-2026
- MICCAI: Medical Image Computing and Computer Assisted Intervention 2023-2026
- IJCAI: International Joint Conference on Artificial Intelligence 2024-2026
- WACV: The IEEE/CVF Winter Conference on Applications of Computer Vision 2025-2026
- ICCV: International Conference on Computer Vision 2025
- ICML: International Conference on Machine Learning 2025
- ECCV: European Conference on Computer Vision 2026
- ICLR: International Conference on Learning Representations 2026
- AISTATS: International Conference on Artificial Intelligence and Statistics 2025-2026
- ECAI: European Conference on Artificial Intelligence 2025
- AAAI: The Association for the Advancement of Artificial Intelligence 2026

Journal Reviewer

- Medical Physics 2024-2025
- IEEE Transactions on Medical Imaging (TMI) 2023-2025
- Transactions on Machine Learning Research (TMLR) 2024-2025
- Computers in Biology and Medicine 2025
- Journal of Medical Imaging 2024
- Smart Health 2023
- Scientific Reports – Nature 2021

Invited Talks

- CS Seminar: Medical Image Segmentation with Transformers, Wayne State University 2024
- BBC Virtual Seminar, Karmanos Cancer Institute, Wayne State University 2024